ÇALIŞMA SAYFASI/STUDY PAGE

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Yapılan İş/Work Accomplished	**Tarih/Date**
Aeronautical Laboratory - COMSOL and Plane Project	**01.02.2024 - 12.02.2024**

I started working as an intern at the Aeronautical Laboratory.I learned the COMSOL application, which is generally used in the field of materials science and has the capacity to perform important simulations for aircraft. While learning the application, I preferred to master it by practicing. I learned the application by completing the tasks given by my supervisor. These are the tasks I completed in 12 day:

- *Determining the speed of any structure I have drawn in the application at any point in a turbulent flow environment has a linear elastic character under the influence of gravity by fixing it from the stationary point of any 3D substance.*
- *To create a mesh analysis of the structure and understand the character of the structure by assigning material to each part of the aircraft wing profile, giving thickness and offset, and using shell analysis.*
- *Confirming the pointwise velocity values and turbulence area of the structure by performing mesh analysis of an airfoil sample taken from the NACA airfoil designator in the air environment.*
- *Confirming the behavior of turbulent flow in the air environment by drawing a structure in the form of an airplane cowl.*
- *Simultaneously simulating the material that this pin is made of and under which loads it will deformed by drawing a pin that is used to connect two elements in airplanes.*
- *Simultaneous monitoring of the durability of a copper pole attached to a wall vertically, depending on certain strength coefficient and temperature under stress.*
- *Simulating how much a temperature block can cool a heated transistor, at what values, by assigning it to the device with active and passive cooling options.*

12.02.2024

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(Tarih, Mühür ve İmza)
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Yapılan İş/Work Accomplished	**Tarih/Date**
Aeronautical Laboratory - COMSOL and Plane Project	**13.02.2024 - 23.02.2024**

```
#include <Wire.h>
#include <Servo.h>
#include <MPU6050.h>

Servo servo;
MPU6050 mpu;

int servoPosition = 83; // Başlangıç pozisyonu
float targetAngle = 0.0; // Hedef açı
float rollAngle = 0.0; // Roll açısı
bool isCalibrated = false; // Kalibrasyon durumu

void setup() {
  servo.attach(9); // Servo motorun sinyal pini 9. pine bağlanır.
Wire.begin();
mpu.initialize();

Serial.begin(9600); // Seri iletişim başlatılıyor

/* MPU6050 kalibrasyonu
calibrateMPU6050();*/

  // Uçağı hareket etmesi için bir tuşa basılmasını bekleyin
Serial.println("Uçağı sabit bir yüzeye koyun ve bir tuşa basın.");
while (!Serial.available()) {
// Bekle
  }
Serial.read(); // Beklemek için kullanılan tuşu temizle

  // Kalibrasyon tamamlandı
  isCalibrated = true;
  Serial.println("Kalibrasyon tamamlandı. Uçak hareket edebilir.");
}

void loop() {
  if (!isCalibrated) {
    // Kalibrasyon tamamlanana kadar bekle
    return;
  }

  // MPU6050'den verileri alın
  int16_t ax, ay, az, gx, gy, gz;
  mpu.getMotion6(&ax, &ay, &az, &gx, &gy, &gz);

  // Roll açısını hesaplayın (derece cinsinden)
  rollAngle = atan2(ay, az) * (180.0 / PI);

  // Hedef açıyı belirleyin (örneğin, 0 derece olarak)
  if (rollAngle > 0 & rollAngle < 5) {
    targetAngle = 30.0; // Sağa dönme
  } else if (rollAngle < 0 & rollAngle > -5) {
    targetAngle = 145.0; // Sola dönme
  } else if (rollAngle < -5.0 & rollAngle > -10) {
    targetAngle = 160.0; // Sola dönme
  } else if (rollAngle > 5 & rollAngle < 10) {
    targetAngle = 45.0; // Sağa dönme
  } else {
    targetAngle = 83.0; // Normal pozisyon
  }

  // Servo motoru hedef açıya döndürün
  servo.write(targetAngle);

  // Değerleri seri monitöre yazdır
  Serial.print("X ivmelenme: ");
  Serial.println(ax);
  Serial.print("Y ivmelenme: ");
  Serial.println(ay);
  Serial.print("Z ivmelenme: ");
  Serial.println(az);
  Serial.print("Açısal Hız (gyro): ");
  Serial.println(gy);
  Serial.print("Roll Açısı: ");
  Serial.println(rollAngle);

  delay(10); // Döngüyü biraz yavaşlatmak için
}

/*void calibrateMPU6050() {
  // Uçağı düz bir yüzeye koyun
  // Kalibrasyon süreci
  // Kalibrasyon işlemleri burada gerçekleştirilir
  // Uçağı hareket ettirmeyin

  Serial.println("Kalibrasyon tamamlandı.");
}*/
```

The algorithm is simply like this. Apart from this, neural network methods and machine learning will be used to optimize the path that will enable the aircraft to fly in a way that reduces its speed at least. This will be used with ESP32 or another controller board in the future. I also learned about the weight point of the plane on the carbon fiber rod with XFLR5 and placed the components accordingly.

23.02.2024

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Yapılan İş/Work Accomplished	**Tarih/Date**
Aeronautical Laboratory - COMSOL and Plane Project	**13.02.2024 - 23.02.2024**

After learning the COMSOL, I started to learn the XLFR5 program. I working these 8 days by generally on XFLR5 and trying this application on projects like I did in the COMSOL application.

XFLR5 is an analysis tool for airfoils, wings and planes operating at low Reynolds Numbers. It includes:

- *XFoil's Direct and Inverse analysis capabilities.*
- *Wing design and analysis capabilities based on the Lifiting Line Theory, on the Vortex Lattice Method, and on a 3D Panel Method.*

Meanwhile, I started to do a project of my own. This aircraft consists of Naca 4412 airfoil design , avionics compartment, carbon fiber rod and servo motor.

Since I want to keep the aircraft at as low weights as possible, the airfoil consists of strong Corafoam. All components of the aircraft are connected to the carbon fiber rod. The avionics part made of 3D Low Weight PLA.

I tried to make the fuselage of the plane from light-density wood, but it is very difficult to cut such low-density wood with a laser cutter. While creating space in the wood to reduce the weight of the wood, the wood burns.

If we look at the avionics part, it consists of gyroscope, inclinometer and Arduino Nano. The status of the x, y and z axes received from the gyroscope is processed in Arduino nano along with the speed and height data received from the inclinometer. In the code I wrote, the logic is simply this: When we launch the plane forward from a high place, it needs to perform maneuvers to ensure that there is always stay in same feet with the data it receives. It performs these maneuvers by sending signals to the servo motors to which the wings are connected.

23.02.2024

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Yapılan İş/Work Accomplished	**Tarih/Date**
Aeronautical Laboratory - Quadcopter Project	**26.02.2024 - 04.03.2024**

We are planned to build a large quadcopter. First of all, I created the shape of the drone using SolidWorks. Afterwards, we decided how the final version of the drone would be by combining all the components on the drone on SolidWorks. After that, we made simulations to measure the durability of the drone via COMSOL. As a result, we predicted that the drone could carry up to 6 kilos without any problems.

The drone has 4 motors with 632 watts of power that can operate between 3-phase BLDC 24V-48V. To drive these motors, I used 4 3-phase ESCs with a 40 amp rating. We used a 25 volt 15000 mAh Lithium polymer battery and used a multi-input boost converter to increase this voltage. Pixhawk orange had used. The Cube autopilot is a further evolution of the Pixhawk autopilot. It is designed for commercial systems and manufacturers who wish to fully integrate an autopilot into their system. In order to measure the current coming out of the motors, a current reading sensor was placed between the ESC input and output and was connected to the pixhawk to control the current. GPS Here3, which includes modules such as gyroscope, inclinometer and caliper, and Herelink antenna system, which allows us to control it with a remote control, were used. Finally, wrote code for the camera with a sensor to enable object recognition with the camera, and it became an object recognition-based camera.

04.03.2024

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Yapılan İş/Work Accomplished	**Tarih/Date**
Aeronautical Laboratory - Electronic Speed Controller	**05.03.2024 - 14.03.2024**

At the request of my supervisor. I started making a 3 phase BLDC Motor Controller circuit.

Since the motor is rated at 2500 watts nominal, the motor driver is rated for nominal 48 V operation. It is designed to operate at voltage and nominal operating current of 50 A. This value is It is a value that meets your needs. At the same time, the instantaneous voltage experienced by the motor driver and current values are also taken into account. A panel is designed to withstand instantaneous 100A. designed. If you change the Mosfets used on the card, the power value that the card can handle will also change.

For ease of production and space saving, all circuit elements except Mosfet blocks are placed on the surface. They are used as assembled (SMD) circuit elements.

In capacitor selection, the maximum voltage value of the usage area is the minimum voltage value of the usage area. capacity requirement, operating temperature and instantaneous large current are selected. Failure of capacitor In order to avoid any problems in case of It was decided to connect the capacitor in parallel. Watt values according to the current passing through the resistors, resistance value according to datasheet data The selection was chosen with low tolerance in the resistors used as reference.

When choosing MOSFETs, the maximum voltage and current values that the motor driver will experience should be taken into account. taken. The safety coefficient is taken as 2 when choosing MOSFETs against sudden current increases. Same Mosfet's internal resistance price-performance criteria to protect motor driver efficiency and heat at the same time was chosen as low as possible.

14.03.2024

ONAY/ CONFIRMATION
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Yapılan İş/Work Accomplished	**Tarih/Date**
Aeronautical Laboratory - Electronic Speed Controller	**14.03.2024 - 29.03.2024**

For use in motor drivers IXTH200N10T N-Channel mosfet was selected. This mosfet high continuous switching current has values. Under 200A continuous current while it can operate at instantaneous peak values of 500A It can also work. Engine in control phase encounter any excessive power consumption.In this case, these mosfets will be able to withstand. It has a voltage value of 100V.

IRS21867 is used as Mosfet driver. both high tier and low tier It can drive MOSFETs and provide relatively high voltage outputs such as 4A. It was preferred because it can work in currents. It can provide values between 6.8V and 20V.Compared to similar products on the market It has a normal switching time of 170ns.

The NCV33035 requires motor speed data to operate as a closed loop. Because In addition to the NCV33035 control integrated, it processes the data received from the position sensor and converts it into speed data. MC33039DR2G integrated circuit is used, which converts and provides feedback to the control integrated.

*There is a 3% efficiency loss due to the MOSFET blocks on the circuit and the driver integrated circuit. is formed. The reason for this loss of efficiency is the internal resistance of the mosfet at high currents. is the resistance value. Therefore, it is estimated that there will be a loss of 40 Watts in the mosfet. this is lost It comes out of the integrated circuits as heat. Apart from this, even when the mosfet is off, it is negligible. It can transmit some. Since the motor is nominally designed as 48V and 50A, the power is calculated as 2400 Watt from P=V.I equation. acceptable. At this current value, there will be a loss of 40 Watts due to MOSFETs. This is a percentage As (40/2400) *100, approximately 2% loss occurs only from MOSFETs.*

29.03.2024

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